

# Solving More Equations by Factoring (Homework)

Solve.

Answers

$$\textcircled{1} b^3 + 5b^2 + 6b = 0$$

$$b = 0, -2, -3$$

$$\textcircled{2} 5y^3 - 20y = 0$$

$$y = 0, 2, -2$$

$$\textcircled{3} -4b^5 + 28b^4 - 24b^3 = 0$$

$$b = 0, 6, 1$$

$$\textcircled{4} 0 = -7y^3 + 7y$$

$$y = 0, 1, -1$$

$$\textcircled{5} 0 = 10x^{10} - 10x^9 - 120x^8$$

$$x = 0, 4, -3$$

$$\textcircled{6} y^3 + 3y^2 + 3y + 9 = 0$$

$$y = -3$$

$$\textcircled{7} 8b^7 - 128b^5 = 0$$

$$b = 0, 4, -4$$

$$\textcircled{8} 0 = y^5 - 16y$$

$$y = 0, 2, -2$$

$$\textcircled{9} -36y^2 + 54y = -6y^3$$

$$y = 0, 3$$

$$\textcircled{10} x^3 - 2x^2 = 4 - 2x$$

$$x = 2$$

$$\textcircled{11} -2b^7 - 8b^5 + 10b^3 = 0$$

$$b = 1, -1, 0$$

$$\textcircled{12} 0 = 2x^{11} + 3x^{10} - 9x^9$$

$$x = 0, \frac{3}{2}, -3$$

$$\textcircled{13} -2b^6 - 4b^2 = 6b^4$$

$$b = 0$$

$$\textcircled{14} 4x^5 = 4x^3 + 48x$$

$$x = 0, 2, -2$$

$$\textcircled{15} 3x^4 + 9x^3 = 30x^2$$

$$x = 2, -5, 0$$

$$\textcircled{16} 5x^2 = 45$$

$$x = 3, -3$$

$$\textcircled{17} 90x^5 - 1000x^3 = 0$$

$$x = 0, \frac{10}{3}, -\frac{10}{3}$$

$$\textcircled{18} -6x^9 + x^8 = -12x^7$$

$$x = 0, -\frac{4}{3}, \frac{3}{2}$$

$$\textcircled{19} 6x^3 = 60x - 26x^2$$

$$x = -6, \frac{5}{3}, 0$$

$$\textcircled{20} x^5 + x + 2 = -2x^4$$

$$x = -2$$

$$\textcircled{21} 8y^3 + 2y^2 = y$$

$$y = 0, \frac{1}{4}, -\frac{1}{2}$$

$$\textcircled{22} 3b^5 - 27b = 24b^3$$

$$b = 0, 3, -3$$